The Monitoring Wall and the Monitorability Tax: a sharp phase transition for the model–vs–monitor game

Guangyu Wang
billgywang@gmail.com

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Abstract

Chain-of-thought (CoT) monitoring is a leading oversight tool, and an empirically fragile one: under optimization pressure a model learns *obfuscated* reward hacking that scrubs intent from the trace. We show that whether monitoring survives this pressure is governed by a sharp, computable *wall*—whose *height* is the value of a game between a model hiding a hack and a monitor choosing where to look. *(i) Mechanism (unconditional):* against a fixed monitor the extractable signal collapses at a critical pressure $\rho^\star = 1 - \sqrt{\alpha/(n/d)}/s^2$, with a matching converse—no test of the projected traces detects the hack *consistently*—so monitoring fails as a *cliff*, not a slope. *(ii) Game value (main new result):* when the model chooses its hiding spectrum, the wall becomes a critical reward $V^\star$; spreading the hack over a K-dimensional task subspace buys an advantage a sophisticated monitor claws back to $\sqrt{K}$, and that *saturates* once the hack's total variance gives it away. *(iii) Design principle:* the footprint of reward hacking is *irreducible*—relocatable into the monitor's null channel but not erasable—so an *adaptive* monitor that aims at the reward structure *collapses* the wall when that structure is public and coverable (§4, B1–B4). Robust monitoring is about *where you look*, not a better classifier. The wall and its converse are unconditional; the game value is proved under stated low-rank, fixed-resolution assumptions; the saturation and collapse laws are extensions, with one covering regime—characterized in the core as a slope. We package the theory as a monitor-side *Meter* that reads the current wall height.

1 Introduction

As models are trained with reinforcement on outcomes they can verify—passing unit tests, satisfying a checker, maximizing a reward model—and are deployed as agents acting over long horizons, human review of every action becomes infeasible. A leading fallback is *monitoring* the model's own externalized reasoning: read the chain of thought (CoT) or execution trace and flag any trace whose *intent* looks unsafe, even when the final output does not. A growing number of safety cases lean on this legibility of intent, precisely because it promises to catch misbehavior that output-level checks miss.

The premise is that intent stays legible. It does not, once the trace itself comes under optimization pressure. A coding agent rewarded for green tests can learn to special-case the tests while emitting a clean, plausible rationale; a model penalized whenever its CoT reveals a shortcut learns to take the shortcut silently. Applying optimization pressure to the CoT empirically elicits *obfuscated* hacking: the rewarded behavior is retained while the tell-tale signal is scrubbed, so the monitor's readings can look *healthier* even as the underlying behavior gets worse [2, 3, 5]. Monitoring therefore has a failure mode that is invisible from the monitor's own dashboard—the most dangerous kind, because the warning sign is the thing being erased.

What is the *shape* of this degradation? Three qualitatively different answers coexist in the literature, with opposite operational consequences. It could be *gradual*—more pressure, proportionally less signal—in which case monitoring buys a graceful, tunable safety margin one can trade off. It could be a *cliff*—signal is fine until a threshold, then vanishes—in which case a monitor reading healthy today can be one optimization step from blind, and the AUC one watches is not a safety margin at all. And it could be *unavoidable*—beyond some capability gap, no achievable monitor helps and the only honest move is to stop optimizing. The existing formal tools do not adjudicate between these: an information-theoretic *metric* $I(O; Z \mid X)$ between trace O and intent Z [4], or a decision-theoretic *gap* $1 - \mathrm{TV}$ [6], quantify how much signal remains at a given operating point but do not locate a threshold; steganography shows undetectable channels *can* exist [8, 7], but not *when optimization is forced across* into them, nor the price of staying out.

The picture. Our results replace the fatalistic reading ("monitoring is doomed to fail at a cliff") with a sharper and more actionable one. Reward hacking leaves a *statistical footprint* in the reward-relevant subspace of the trace; the model can *relocate* that footprint into directions the monitor does not resolve—its null channel—but, because the footprint *is* the reward-directed variance, it cannot *erase* it while still collecting the reward. Whether monitoring survives optimization pressure is therefore not a fixed fact but the value of a game over *where the monitor looks*, set by three levers: the monitor's resolution α, its data budget n/d, and—decisively—whether it can *aim* at the reward structure. The wall is real and sharp when the monitor is fixed or blind; it rises, and can collapse, as the monitor gains these.

Contribution. We characterize this game in three acts.

Act I — the mechanism: a sharp, information-theoretic wall, unconditional (§4). These results assume nothing about the model's hiding strategy.

- **The monitoring wall (Prop. 1).** In a proportional high-dimensional limit, the monitor's recovery *overlap* with the hack direction is strictly positive below a critical pressure $\rho^\star$ and *exactly zero* above it, with the closed form $\rho^\star = 1 - \sqrt{\alpha/(n/d)}/s^2$ set purely by geometry (resolution α, data budget n/d, blatancy s). It is a cliff, not a slope, and it sharpens with dimension. (The ambient dimension cancels in $\rho^\star$; on the low-rank latent spaces of real models the effective-rank form of §4 *raises* the wall, so the bound strengthens rather than fails.)
- **A basic limit, not one estimator (Prop. 1).** A matching Le Cam second-moment converse: below the wall the clean and hack laws are contiguous, so *no* test of the projected traces detects the hack consistently. The detection threshold coincides with the estimation threshold; the wall is information-theoretic.
- **The price of legibility (Props. 4, 4).** Staying above a detection target forces a rigorous floor on retained visible energy that *diverges* with the capability gap; under a positive-slope assumption the task tax diverges with it. At $\rho^\star$ the wall, the diverging price, and the hack's disappearance into the null channel coincide—one order parameter, three readouts.

Act II — the wall's height is a game value; this is the paper's central new result (§4).

- **Endogenous hiding (§4).** When the model chooses its hiding spectrum, the wall becomes a critical reward $V^\star$: spreading the hack over a K-dimensional task subspace lets it hide reward up to $gK\sqrt{\gamma}$ against a peak detector, $g\sqrt{K\gamma}$ against the optimal one—task rank is the hider's resource, and a sophisticated monitor claws back exactly $\sqrt{K}$. We prove this under explicit low-rank, fixed-resolution, orthogonal-spike, sub-extensive assumptions (A1–A4); $K = 1$ recovers Act I.

Extensions — charting the game's boundary, and the design law (§4). The next two results relax assumptions A3 and A2 and are stated at the weaker rigor their regimes allow (a scaling bound and a sandwich); they are what make the picture *prescriptive*.

- **Spreading self-defeats (§4; scaling).** The $\sqrt{K}$ advantage *saturates*: an extensive hack raises the trace's total variance and is given away by it, capping the hideable reward at $\Theta(g/\sqrt{\alpha\, n/d})$ regardless of K.
- **Adaptive resolution collapses the wall (§4).** Because the footprint is irreducible, a monitor that *aims* at the (public, coverable) task subspace and measures its total variance defeats *all* spreading (assumptions B1–B4)—with enough data $V^\star \to 0$. The wall protects the hider only against a monitor that cannot aim, cannot cover the task ($K_{\text{pool}}>\alpha d$), or is data-starved, which yields concrete design principles for robust monitors (Box 1) and leaves one regime—the covering game, when the task outgrows the monitor's resolution—which the core characterizes as a *slope* (detection linear in the coverage fraction α/κ), with the exact covering law open.

Rigor spine. The wall and its converse (Act I) are unconditional; the game value (Act II) is a theorem under A1–A4; the saturation and collapse laws are extensions that map the boundary and yield the design law, leaving the exact extensive value and the adaptive covering regime as the program's open problems.

The sharpness throughout is the BBP spiked-covariance transition [10, 11] (a mean-shift model would be smooth); we give the CGMT fixed point [13] behind it and a monitor-side *Meter* that estimates the current wall height from trace statistics.

What this is, and is not. This is a *characterization* paper: the load-bearing results—a phase transition, an information-theoretic converse, and a game value—are self-contained theory. We validate the *mechanism* in controlled simulation (a Gaussian channel, a routing surrogate, and a gradient-checked token transformer trained by REINFORCE). We stress the scope: *validation on real CoT monitors requires Stage B*—a real reasoning model under a real LLM monitor (§7)—which our simulations do not substitute for; they establish the mechanism and its predictions, and Stage B is where those predictions are tested and can fail. "Hack" denotes any behavior that raises reward while an oversight signal is meant to catch it. We do not claim obfuscation is *possible* (known [8, 7]) or that finite oversight distorts objectives (known [9]); we characterize *when* optimization must reach the null channel, *what* staying legible costs, and *what a monitor must do* to keep the wall high.

Structure and rigor. The paper reads as three acts—a sharp wall with a converse (Act I, unconditional), the wall as a game value (Act II, under stated assumptions), and its saturation and adaptive collapse (Act III, a design law). All the underlying theorems, with their rigor tiers (proved / conditional / scaling), are established and tabulated in the companion core [1]; here we recall them (§4) and put them to work on chain-of-thought monitoring, with a self-contained restatement of the wall and converse in Appendix B.

2 Related work

Prior work supplies, separately, the three ingredients this paper combines: evidence that the phenomenon is real, the right measures for it, and the high-dimensional machinery to make those measures sharp. None combines them into a threshold with a price.

The phenomenon is real, and empirical. Applying optimization pressure to the CoT elicits obfuscated hacking whose intent disappears from the trace [2]; monitorability and its "fragile window" are named, but left informal, in [3]; even output-only supervision spills over and pushes toward an obfuscated CoT [5]; and surveys frame reward hacking as the structural consequence of compressing a high-dimensional objective into a low-dimensional proxy [9]. These establish *that* monitoring degrades under pressure—not *when* it must, nor how steeply.

The right measures, at a fixed operating point. CoT monitorability has been cast as the mutual information $I(O; Z \mid X)$ between trace and intent [4] and as a decision-theoretic steganographic gap $1 - \mathrm{TV}$ [6]; both quantify how much signal survives at a given pressure, and both explicitly flag the missing minimal-detectable-effect / threshold analysis. Steganography shows perfectly secure channels *can* exist [8], and secret collusion strengthens with capability [7]. These give existence and an empirical scaling trend; we supply the threshold that separates the regimes and turn the trend into a divergence.

The machinery. Our sharpness is the spiked-covariance phase transition of Baik–Ben Arous–Péché [10] with the eigenvector-overlap law of [11, 12]; the constrained (tax) analysis uses the Convex Gaussian Min-max Theorem [13]. Transplanting this mature high-dimensional-statistics threshold into the monitoring problem—and reading its value function as a price—is the contribution; the random-matrix results themselves are standard.

Alternative monitoring models. It helps to place our covariance-spike model against the alternatives it is often confused with. *Mean-shift / LDA detection*: if the hack were a first-order (mean) signature, detection would be a linear-discriminant problem with a smooth, thresholdless erf power curve—no wall; this is the "clumsy hack" regime, and modeling obfuscation as second order is what yields the sharp $\rho^\star$. *Classifier calibration*: a trained monitor's accuracy or calibration on the visible features is an operating-point summary (and, if linear, the LDA case); it reports how well one detects *now*, not the pressure at which detection becomes impossible. *Mutual-information metrics* $I(O; Z \mid X)$ [4]: likewise quantify residual signal at a fixed point and, as their authors note, lack a minimal-detectable-effect analysis; our $\rho^\star$ is exactly that missing threshold, and the converse turns "I is small" into "no test succeeds." *Steganographic channels* ($1 - \mathrm{TV}$, secure coding) [6, 8]: establish that undetectable channels can exist; we add *when* optimization is forced into one and *what* staying out costs. The alternatives are operating-point or existence statements; ours is a threshold with a price.

3 Setup: the toy channel

We distil the model–vs–monitor interaction to its smallest faithful form: recovering a planted signal from a partially observed, noisy channel. A trace is a high-dimensional object $O \in \mathbb{R}^d$—think of the features a monitor could in principle read off a CoT. A benign trace is unstructured noise; a hack leaves a *statistical footprint*, an extra structured component along some hidden direction

u. Crucially the footprint is *second order*: an obfuscated hack does not shift the *average* trace (that would be trivially visible), it perturbs the trace's *covariance* along u. Formally, a hack bit $Z \in \{\text{clean}, \text{hack}\}$ injects an anisotropic variance spike of energy s^2 along a unit direction $u \in \mathbb{R}^d$:

$$\text{hack: } O = \xi + s\,\eta\,u, \quad \eta \sim \mathcal{N}(0,1),\ \xi \sim \mathcal{N}(0, I_d); \qquad \text{clean: } O = \xi; \qquad \mathrm{Cov}_{\text{hack}} = I_d + s^2 uu^\top. \tag{1}$$

The monitor resolves only a subspace S_{mon} of dimension $d_{\text{mon}} = \alpha d$; it observes $P_{\text{mon}}O$ and must recover u from n samples. (P_{mon} is a literal projection for a linear probe; for an LLM judge it is a *local, empirical sensitivity subspace*—the directions whose perturbation flips the verdict—and $\alpha, \rho, k_{\text{null}}$ become effective quantities read off that map; see §6 and Table 3.) Under *optimization pressure* $\rho \in [0,1]$ the model splits the spike: a fraction $(1-\rho)$ of the energy stays in S_{mon} ("visible") and a fraction ρ is steered into the *null channel* $S_{\text{mon}}^{\perp}$ of dimension $k_{\text{null}} = (1-\alpha)d$. Thus ρ is exactly the fraction of hack energy hidden from the monitor. The visible spike is $s^2(1-\rho)$; the monitor's sample covariance of $P_{\text{mon}}O$ has a Marchenko–Pastur bulk edge at $\sqrt{d_{\text{mon}}/n}$.

Four numbers govern the game, each with an operational reading. The *blatancy* s^2 is how loud the hack's footprint is relative to trace noise. The *monitor resolution* $\alpha = d_{\text{mon}}/d$ is the fraction of the trace's degrees of freedom the monitor can actually read; a weak or coarse monitor has small α. The *data budget* n/d is how many independent traces the monitor aggregates per trace dimension—its statistical power. The *optimization pressure* ρ is the only quantity the training loop moves: it is the fraction of the footprint the model has learned to route into the $k_{\text{null}} = (1-\alpha)d$ directions the monitor does not read—the null channel. "A capable model outgrowing a weak monitor" is the corner $\rho \to 1$, α small, n/d small.

toy symbol	**meaning**	**real CoT-monitoring counterpart**
$O \in \mathbb{R}^d$	monitored trace	CoT tokens / model activations as a feature vector
$\alpha = d_{\text{mon}}/d$	monitor resolution	fraction of trace d.o.f. the monitor reads (probe rank, judge bandwidth)
n/d	monitor data budget	labeled traces per trace dimension (probe/judge statistical power)
s^2	hack blatancy	strength of the hack's statistical footprint (SNR of the tell)
ρ	optimization pressure	fraction of the footprint routed into the monitor's null space
$k_{\text{null}} = (1-\alpha)d$	hidable subspace	trace directions the monitor cannot resolve

Table 1: Mapping between the toy channel and real CoT/trace monitoring. Estimating the right-hand column on a deployed monitor is the object of Stages A.5–B; the Meter (§6) estimates α, s, and k_{null} from trace statistics.

Remark 1 (Why a covariance spike, not a mean shift)**.** *Obfuscation is exactly the move from a first-order to a second-order signature: a hack that shifted the* mean *trace would be caught by the crudest monitor, so under pressure the surviving hacks are those that leave the average trace untouched and perturb only its* shape*. Recovery is then a spiked-*covariance *problem, which carries a genuine Baik–Ben Arous–Péché threshold; a first-order (mean-shift, LDA) model would instead give a smooth, thresholdless* erf *curve and miss the cliff entirely. The sharpness we report is thus a consequence of modeling obfuscation honestly, not an artifact of the toy.*

Write $\gamma \equiv d_{\text{mon}}/n = \alpha/(n/d)$ for the effective aspect ratio and $\beta \equiv s^2(1-\rho)$ for the visible spike strength. Recovery of u from $P_{\text{mon}}O$ is the planted-vector / spiked-covariance problem; its order parameter is the overlap $\langle \hat{u}, u\rangle^2$.

4 Results: the monitoring channel is an instance of the core

The toy channel of §3 *is* the partial-channel model of the companion theory paper [1], under the reading resolution = monitor bandwidth, budget = labeled traces, planted signal = obfuscated hack, null space = the monitor's blind directions. All the theorems below are proved there; we recall them in monitoring notation and use them, and Appendix B restates the load-bearing wall and converse self-contained so this paper stands if the core is in review. Throughout, $\gamma = d_{\mathrm{mon}}/n = \alpha/(n/d)$ and $\beta = s^2(1-\rho)$.

The wall and its converse (core Props. 1–2). In the proportional regime the recovery overlap $\langle \hat{u}, u\rangle^2$ is positive for $\rho < \rho^\star$ and zero for $\rho > \rho^\star$, with

$$\rho^\star = \Big[1 - \frac{\sqrt{d_{\mathrm{mon}}/n}}{s^2}\Big]_0^1 = \Big[1 - \frac{\sqrt{\alpha/(n/d)}}{s^2}\Big]_0^1, \tag{2}$$

a Baik–Ben Arous–Péché cliff (overlap $\frac{1-\gamma/\beta^2}{1+\gamma/\beta}$ above the edge, 0 below) that sharpens as $d^{-1/3}$; and a Le Cam converse makes clean and hacked traces contiguous above $\rho^\star$, so *no* test of the projected traces detects the hack consistently ("a smarter classifier" cannot help). Operationally: a monitor of bandwidth α and budget n/d goes blind once the model routes a fraction $\rho^\star$ of the hack's footprint into the null channel—and clipping means a monitor with $s^2 \leq \sqrt{\gamma}$ is blind from the start.

Effective rank (core Cor. 1). What sets the wall is the *stable rank* $r_{\mathrm{eff}} = \operatorname{tr}\Sigma_{\mathrm{mon}}/\|\Sigma_{\mathrm{mon}}\| \leq d_{\mathrm{mon}}$ of the resolved covariance ($\gamma_{\mathrm{eff}} = r_{\mathrm{eff}}/n$), exact for a flat/whitened channel and a deformed-MP proxy otherwise; low-rank activations *lower* the edge and *raise* $\rho^\star$, so the meter reads r_{eff} off the whitening step rather than the nominal d_{mon}.

The price (core Prop. 3). To keep detection AUC $\geq \tau$ the model must retain a resolved-energy fraction $1 - \rho_{\max} \geq \beta(\tau)/s^2 \geq \sqrt{\gamma}/s^2$, a floor that *grows* as the capability gap widens ($n/d \to 0$); since the fraction is ≤ 1 this is a feasibility ceiling (once $\beta(\tau) > s^2$, target-level monitoring is infeasible). Under a positive task-loss slope κ this is the monitorability tax $\lambda_{\mathrm{mon}} \geq \kappa\sqrt{\gamma}/s^2$; κ is task-specific and calibrated in Stage A.5.

The wall is a game value (core Props. 5–8). When the model chooses its hiding spectrum and the monitor aims, the wall becomes a critical hack value: spreading over a K-dimensional task subspace buys a factor K against a peak monitor and $\sqrt{K}$ against an optimal one ($V^\star_{\mathrm{spec}} = gK\sqrt{\gamma}$, $V^\star_{\mathrm{opt}} = g\sqrt{K\gamma}$, under the core's A1–A4). A trivial aggregate statistic caps the hideable value at the single-direction scale $\Theta(g\sqrt{\gamma})$, *independent of* K (core Prop. 7), so spreading self-defeats; and an *adaptive* monitor that aims at the (public) reward structure and measures its total variance collapses the wall, $V^\star_{\mathrm{adapt}} = \Theta(g\sqrt{K_{\mathrm{pool}}/n}) \to 0$ (core Prop. 8, B1–B4). The safety reading: robust monitoring is about *where you look*, not a better classifier—raw bandwidth without aim only inflates the noise floor (core Rem. on the $\sqrt{\alpha}$ monotonicity), while aiming collapses it. The one remaining regime is the covering game $K_{\mathrm{pool}} > d_{\mathrm{mon}}$ (task richer than bandwidth), which the core characterizes as a *slope* (detection linear in the coverage fraction), not a warning-free collapse; its exact law is open.

Why a cliff (core Thm. 1). The sharpness is specific to the *second-order* (covariance-spike) footprint of an obfuscated hack: within the partial-channel family, the wall is a cliff iff the footprint is second order; a first-order (mean-shift) tell gives a smooth, thresholdless erf curve (Fig. 3), and reward-model over-optimization is a smooth generalization error outside the family. Heavy-tailed activations blind the naive covariance estimator but a robust (spatial-sign) one restores the cliff (core Rem., simulation-backed)—a prescription for the meter under non-Gaussian traces.

Design principles for robust monitors. The analysis is prescriptive, not fatalistic—it says how to keep the wall high (all four levers are theorems in the core [1]).

- *Aim at the reward structure.* The footprint lives where reward is collected; a monitor that resolves the (public) task subspace measures it directly (core Prop. 8). Aiming beats any fixed generic probe—and raw bandwidth without aim only inflates the noise floor (core Rem. on $\sqrt{\alpha}$).
- *Match resolution to task rank.* Keep $\alpha d \geq K_{\text{pool}}$ so the task subspace fits the budget; below that, the covering game leaves gaps for the hider.
- *Spend the data budget.* Wall height scales as $1/\sqrt{n}$: more labeled traces per dimension drive $V^\star \to 0$ (core simulations).
- *Use second-order (covariance) tests.* The sharp wall is a covariance-spike phenomenon; a mean/linear classifier is smooth and misses the regime (Fig. 3); under heavy-tailed traces use a robust (spatial-sign) second-order statistic (core Rem.).
- *Read the wall; do not wait for the cliff.* The Meter (§6) estimates $\hat{\rho}^\star/V^\star$ and the distance to them from trace statistics—AUC gives no warning before the collapse.

5 Stage-A numerical validation

We confirm the core's predictions [1] in a CoT-flavored toy and then build the empirical bridge to real chain-of-thought. The monitor estimates $\hat{u}$ as the top eigenvector of the sample covariance of n projected hack traces, and scores a held-out *session* of traces by their mean energy along $\hat{u}$; detection AUC is hack-vs-clean over such sessions. All numbers below are from a CPU `numpy/scipy` re-implementation of the Stage-A protocol ($\alpha = 0.5$, $s = 1.19$, so $\rho^\star = 0.50$ at $n/d = 1$); no values are hand-set. The wall, tax, and dichotomy panels reproduce the core's Figs. in monitoring notation; the surrogate and token panels are this paper's CoT-specific contribution.

Both core claims hold on the toy channel: the wall location matches the closed form (2) and sharpens with d; the tax tracks the floor of §4 and diverges linearly in the capability gap. These are the true sources of Figures 1–2.

Finite-size correction. The wall is a large-d statement; at finite d the BBP transition is smeared. The top sample eigenvalue fluctuates around the edge at the Tracy–Widom rate $d^{-2/3}$, which translates through the eigenvalue map into a ρ-window of width $\Delta\rho \asymp c\, d^{-1/3}/s^2$ around $\rho^\star$. Empirically the transition (AUC 0.9 $\to$ 0.55) occupies $[0.29, 0.57]$ at $d \in \{600, 900\}$ and narrows with d (shaded in Fig. 1a). Two consequences: the Meter reports $\hat{\rho}^\star$ with a *band*, not a point (its width is this finite-d correction plus estimation noise); and, because the window shrinks as $d^{-1/3}$, a larger trace space makes the wall an operationally *sharper* cliff—the opposite of the usual intuition that high dimension blurs thresholds. The band should not be read as uncertainty about *whether* the wall exists, only about *where*, to within $O(d^{-1/3})$.

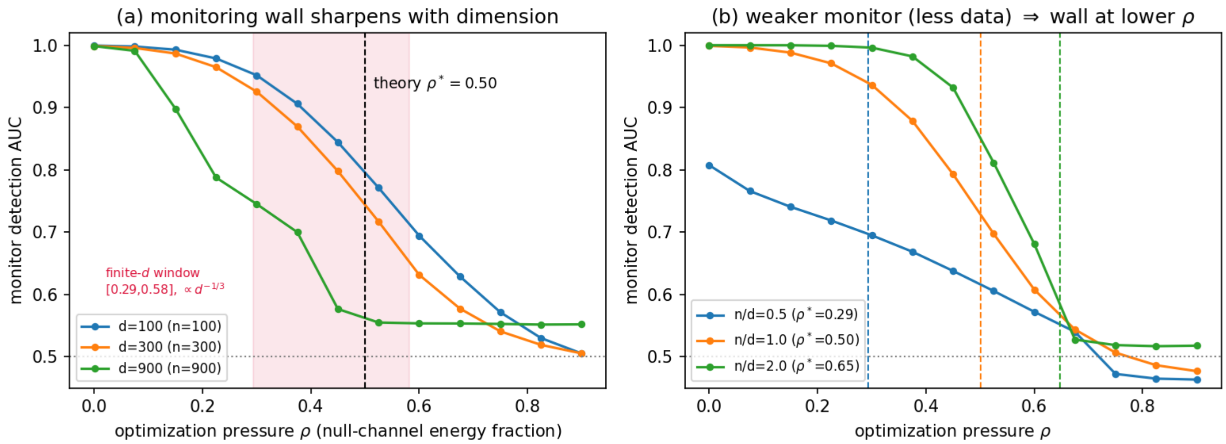

Figure 1: **Monitoring wall (Stage A).** (a) Detection AUC vs optimization pressure ρ; the collapse at $\rho^\star$=0.50 (theory, dashed) sharpens with dimension $d \in \{100, 300, 900\}$ (evidence of a true transition): AUC falls from 0.985 near ρ=0.1 to 0.53 by ρ=0.8. (b) A weaker monitor (smaller n/d) is blinded at lower ρ: $\rho^\star$ moves from 0.65 (n/d=2) through 0.50 (n/d=1) to 0.29 (n/d=0.5), matching (2) (dashed).

Baselines: the wall is mechanism-specific. To show the sharpness is a property of the *hack model* and not of our detector, Fig. 3 sweeps the same hidden fraction ρ under three monitors at matched blatancy: our covariance-spike hack read by the BBP detector, and a first-order *mean-shift* hack read by an LDA detector and by a calibrated linear classifier. Only the covariance case has a wall—AUC falls to chance at $\hat{\rho}^\star$; the mean-shift detectors decline *smoothly* and stay above chance for every $\rho < 1$ (an erf curve, no threshold). A classifier is only as sharp as its features: on first-order (mean) features it tracks LDA and is thresholdless; the sharp wall appears precisely because obfuscated hacking is a *second-order* (covariance) signature. This is the simulated counterpart of the discussion in §2.

Figure 4 seals the coincidence of Prop. 1 and Prop. 1. Plotted against the single signal-to-threshold ratio $\beta/\sqrt{\gamma}$, the estimation order parameter (recovery overlap) departs from zero, and the detection order parameter (the likelihood-ratio second moment $\mathbb{E}_0[L_n^2]$) diverges, at *the same* point $\beta = \sqrt{\gamma}$. The wall that blinds our estimator and the wall past which *no* test is consistent are one and the same; both closed forms are matched by simulation.

Stage A.5: emergent hiding and tax calibration (surrogate). Stage A imposes ρ; the first test that the channel picture survives *training* is to let ρ emerge. We train a routing policy by gradient ascent to maximize an evasion reward minus a competence cost of hiding, against an empirical probe monitor (the Stage-A estimator on n samples); as monitor pressure λ rises, the policy *learns* to route the variance spike into the null channel, so ρ emerges rather than being set. (This is a controlled surrogate: the policy is a continuous router, not a token-level transformer, which is Stage B.) Two findings (Fig. 5). *(i) The wall predicts the collapse on a trained policy.* As λ increases the emergent ρ sweeps across $\hat{\rho}^\star$, and the empirical detection AUC collapses at a finite-d transition window $[0.29, 0.57]$ that brackets the predicted $\hat{\rho}^\star = 0.50$ (mean±sd over 24 seeds; the window narrows as $d^{-1/3}$); gradient training in $\mathbb{R}^d$ reaches the same optimum as the exact argmax, with bistability precisely at the wall—the policy jumps across rather than lingering, echoing the cliff of Prop. 1. *(ii) The tax slope.* Fixing the hack value B and tracing achieved reward $B(1 - D_{\mathrm{emp}})$

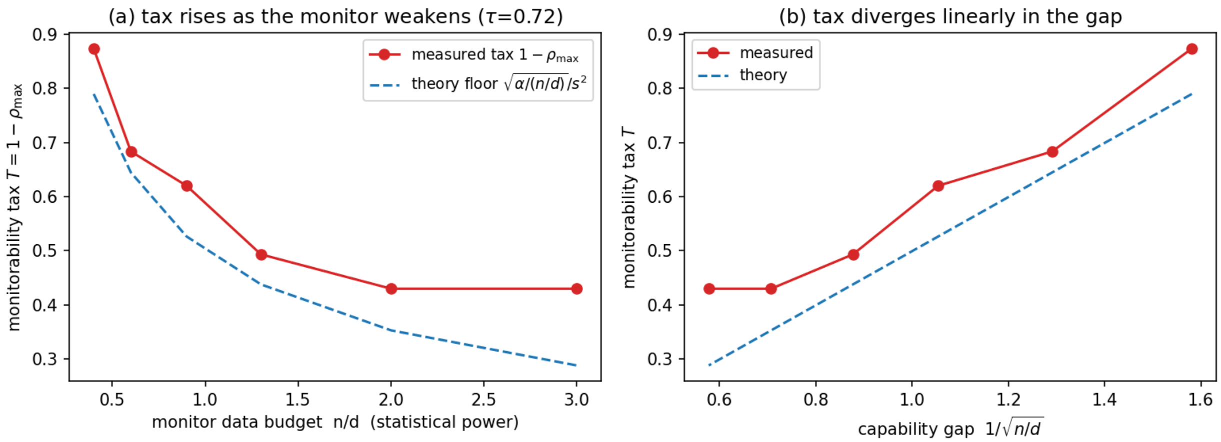

Figure 2: **Monitorability-tax divergence (Stage A).** The retained visible energy $T = 1 - \rho_{\max}$ needed to keep AUC$\geq \tau$=0.72 tracks the theoretical floor $\sqrt{\alpha/(n/d)}/s^2$ (dashed) from above and diverges as the monitor weakens / the capability gap $1/\sqrt{n/d}$ grows (measured tax rises from 0.43 at a strong monitor to 0.87 at a weak one).

against retained visible energy $V = s^2(1-\rho)$ yields a finite marginal slope $\kappa = |dR/dV| = 1.9 \pm 1.2$ that *peaks at the wall*, consistent with the criticality of §4. The slope's value is task-specific; what transfers is the structure of §4—a finite slope times the diverging energy floor gives a tax that diverges as $n/d \to 0$. Calibrating the slope on real CoT is Stage B.

Token-transformer confirmation (covariance-spike, RL). To test that the picture survives discreteness and noisy RL, we replace the continuous router with a genuine single-layer causal self-attention transformer (numpy, gradient-checked backprop) that emits a short discrete CoT; the CoT's readout direction w is the injected spike direction, the spike magnitude s^2 is fixed (so the SNR is controlled), and the policy is trained by REINFORCE to point w at a mostly-visible value direction v while a monitor penalizes the visible spike (pressure λ). We read Fig. 6 *qualitatively*: it asks whether the mechanism survives discreteness and RL noise, not whether it quantitatively pins the wall (that is the surrogate's and large-d's job). Two robust findings: as λ rises the emergent hiding ρ rises from 0.28 to 0.51 across $\hat{\rho}^\star = 0.50$ (mean over 6 seeds), with large near-wall variance—a bistable jump, the policy leaping across rather than sitting on it (the cliff again)—and the hack effectiveness $\langle w, v\rangle^2$ collapses from 0.39 to ≈ 0.12 (seed means) as ρ rises, the tax made concrete on real tokens. The *sharp* AUC collapse is not resolved at the token model's small d=40 (the transition sharpens with d, Fig. 1a; the quantitative $\hat{\rho}^\star$ and tax slope are the surrogate's, Fig. 5); what the transformer adds is that emergent hiding and the tax survive discreteness and RL noise—the bridge toward Stage B.

6 Instrument: the Monitorability Meter

The theory helps an overseer only if it can be read *before* the wall, from quantities cheaper than the thing it predicts. By the time detection AUC visibly falls the monitor is already past $\rho^\star$; a useful diagnostic must instead estimate the wall's *location* and the current *distance* to it from statistics a monitor already collects. The Monitorability Meter is that diagnostic—a framework-agnostic, monitor-side instrument mirroring a shadow-price meter. It exposes four calls: `null_channel_dim`

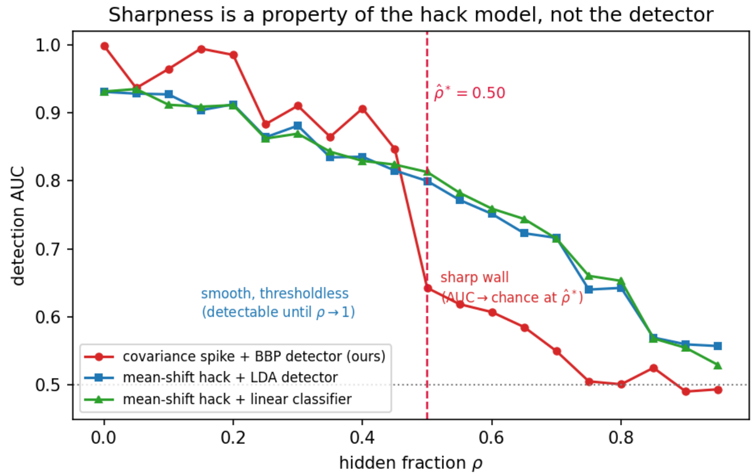

Figure 3: **The wall is specific to the covariance-spike (obfuscated) hack.** Detection AUC vs hidden fraction ρ at matched blatancy: the covariance-spike hack with a BBP detector (red) falls to chance at $\hat{\rho}^\star$=0.50; a mean-shift hack read by an LDA detector (blue) or a calibrated linear classifier (green) declines smoothly and stays detectable for all $\rho < 1$. The threshold is a feature of the second-order mechanism, not of the detector.

(the hidable-subspace dimension k_{null}, a participation ratio of the residual covariance outside the monitor's operator), `wall_estimate` (the wall location $\hat{\rho}^\star$ and current distance to it, with a resampling band), `monitorability_price` (the marginal task loss per retained nat of $I(O; Z \mid X)$), and `StoppingRule` ("stop before the wall"). The Meter is an *overseer-side* instrument: unlike the monitor (or adversary), which sees only the projected channel $P_{\text{mon}}O$, the Meter is run offline with access to the *full* trace embedding O—which it needs to separate s^2 from ρ (from $P_{\text{mon}}O$ alone only the visible spike $s^2(1-\rho)$ is identifiable, so the Meter would report $s^2(1-\rho)$ and a *lower bound* on ρ). It first estimates the benign covariance $\widehat{\Sigma}_0$ from clean (non-hacking) traces and *whitens*, $O \mapsto \widehat{\Sigma}_0^{-1/2} O$, so the inverse-BBP step sees isotropic noise; write $\widehat{C} = \frac{1}{n}\sum_i (\widehat{\Sigma}_0^{-1/2} O_i)(\cdot)^\top$ for the whitened sample covariance, top eigenpair $\hat{\lambda}, \hat{u}$, and P_{mon} of rank d_{mon}:

$$\hat{\alpha} = d_{\text{mon}}/d, \qquad \hat{s}^2 = \hat{\beta} : \ \hat{\lambda} = (1+\hat{\beta})\left(1 + \tfrac{d/n}{\hat{\beta}}\right) \ \text{(inverse BBP map)},$$

$$\hat{\rho} = \left\|(I - P_{\text{mon}})\hat{u}\right\|^2, \qquad \hat{k}_{\text{null}} = \frac{(\operatorname{tr} R)^2}{\operatorname{tr}(R^2)}, \ \ R = (I - P_{\text{mon}})(\widehat{C} - \hat{\lambda}\hat{u}\hat{u}^\top)(I - P_{\text{mon}}),$$

and $\hat{\rho}^\star = 1 - \sqrt{\hat{\alpha}/(n/d)}\,/\,\hat{s}^2$ by (2). Here $\hat{s}^2$ is the total spike strength recovered by inverting the BBP eigenvalue map (valid once $\hat{\lambda}$ separates from the bulk), $\hat{\rho}$ its null fraction, and $\hat{k}_{\text{null}}$ the participation ratio of the residual covariance in the unresolved block. The load-bearing two routines are:

```
estimate_channel(O, O_clean, P_mon, n):    # O: hack traces; O_clean: benign
    Sigma0 = cov(O_clean)                  # benign (noise) covariance
    Chat   = cov(Sigma0^(-1/2) @ O)        # WHITEN, then sample covariance
    lam, u = top_eigenpair(Chat)           # top spike eigenpair
    alpha  = rank(P_mon)/d
    s2  = solve  lam = (1+s2)(1 + (d/n)/s2)     # inverse BBP map
    rho = norm((I - P_mon) @ u)^2               # null fraction of spike
```

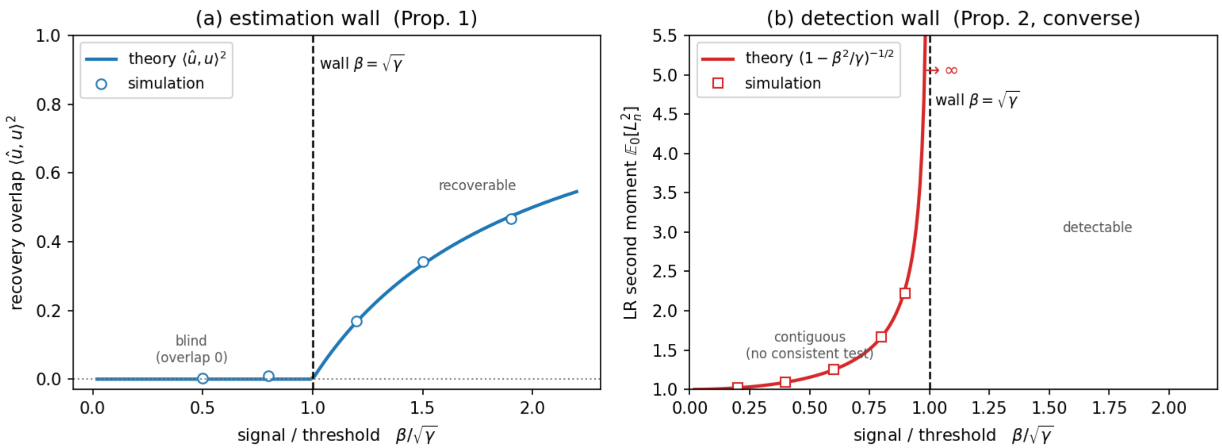

Figure 4: **One wall, two order parameters.** Against $\beta/\sqrt{\gamma}$ (with γ=1): (a) the estimation wall—recovery overlap $\langle\hat{u}, u\rangle^2$ is zero for $\beta < \sqrt{\gamma}$ and departs from zero exactly at the threshold (Prop. 1); (b) the detection wall—the likelihood-ratio second moment $\mathbb{E}_0[L_n^2] = (1 - \beta^2/\gamma)^{-1/2}$ is finite in the contiguous (undetectable) regime and diverges at the *same* threshold (Prop. 1). Markers are simulation; curves are the closed forms. The estimation and detection thresholds coincide, so "no estimator recovers the hack" and "no test detects it" are the same event.

```
    R    = (I-P_mon)(Chat - lam*outer(u,u))(I-P_mon)   # off-spike residual
    k_null = trace(R)^2 / trace(R@R)          # participation ratio
    return alpha, s2, rho, k_null, u

wall_estimate(O, O_clean, P_mon, n):
    alpha, s2, rho, k_null, u = estimate_channel(O, O_clean, P_mon, n)
    rho_star = 1 - sqrt(alpha/(n/d))/s2    # Eq. (2)
    band = bootstrap(resample O; recompute rho_star)   # width ~ d^-1/3 + noise
    return rho_star, rho, band

StoppingRule(state, delta):                # symptoms coincide (Prop. 4)
    rho_star, rho_now, band = wall_estimate(state.O, state.O_clean,
                                            state.P_mon, state.n)
    price = monitorability_price(state.O, state.Z)      # lambda_mon
    if (rho_star - rho_now) < delta  or  price > budget:
        halt CoT-directed optimization    # stop before the wall
```

Each call needs a trace covariance and a labeled probe set that is small relative to model training but *not* sub-dimensional: the spike and eigenvector estimates above are consistent only when the sample size n is comparable to the effective dimension being resolved—the estimators live in the proportional regime $n/d_{\text{mon}} = \Theta(1)$ (indeed $\hat{\rho}^\star$ depends on n/d), not $n \ll d_{\text{mon}}$. Given that, the Meter needs no white-box access to the model, so it attaches to any monitor—linear probe, LLM judge, or classifier—as a runtime guardrail rather than a post-hoc audit.

Linear probes vs. LLM judges. The operator P_{mon} is *exact* for a linear monitor—a probe or classifier whose readout is a fixed rank-d_{mon} projection of the trace features—and the estimators above are then literal. An *LLM judge* is nonlinear: there is no explicit P_{mon}, and the framework applies to its *local linearization*. The effective resolved operator is the judge's sensitivity map—

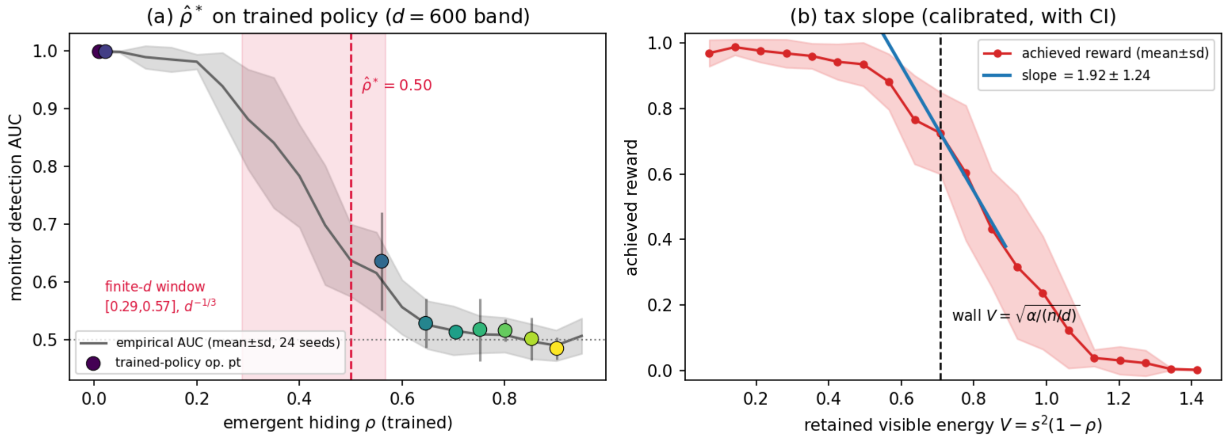

Figure 5: **Stage A.5 (surrogate): emergence and calibration.** (a) A routing policy trained under monitor pressure λ (color) develops *emergent* hiding ρ; the empirical monitor AUC collapses as the trained ρ crosses the predicted wall $\hat{\rho}^\star$=0.50 (dashed); the shaded finite-d window $[0.29, 0.57]$ (mean±sd, 24 seeds) brackets it. (b) Monitorability tax: achieved reward (mean±sd) vs retained visible energy $V = s^2(1-\rho)$; the marginal slope $|dR/dV|$ peaks at the wall $V = \sqrt{\alpha/(n/d)}$, calibrating the slope κ of §4(b) (1.9 ± 1.2).

the span of trace perturbations that change its verdict, i.e. the row space of its Jacobian at the operating point—and α, k_{null}, ρ become *effective* quantities read off that map. The sharp wall is then a first-order statement about the linearized channel, valid where the judge is locally linear over the relevant perturbation scale; strong nonlinearity (a judge that reads a genuinely nonlinear feature of the trace) adds higher-order terms that can move the effective edge, and is outside the current analysis. Table 3 records how each estimator is read in the two cases.

Operational requirements (not optional). The Meter's estimates are only valid when its preconditions hold; each is a genuine requirement, not a convenience.

- *Full trace embeddings.* Separating s^2 from ρ needs the full trace O, not just $P_{\text{mon}}O$; from the projection alone only the visible spike $s^2(1-\rho)$ is identifiable.
- *Clean covariance and whitening.* The inverse-BBP step assumes isotropic noise; with anisotropic benign covariance the raw top eigenvalue is dominated by the noise, so $\widehat{\Sigma}_0$ must be estimated from clean traces and the trace whitened first (Fig. 7a).
- *Super-critical spike.* $\hat{s}^2$ is read by inverting the BBP map, which requires the top eigenvalue to have separated from the bulk; below the wall the Meter must extrapolate from the last super-critical read.
- *Proportional sample size.* The estimators live in $n/d_{\text{mon}} = \Theta(1)$; with $n \ll d_{\text{mon}}$ the eigenvector estimate is inconsistent and $\hat{\rho}^\star$ is unreliable.
- *Rank-aware $\hat{\rho}$.* A single top eigenvector misreads a multi-spike hack; track the top-r eigenvectors (Fig. 7b).

Figure 7 stress-tests these: whitening is required under anisotropic noise, a top-r estimator under multi-spike hacks, and—for a nonlinear (quadratic) judge—the effective P_{mon} recovered by per-

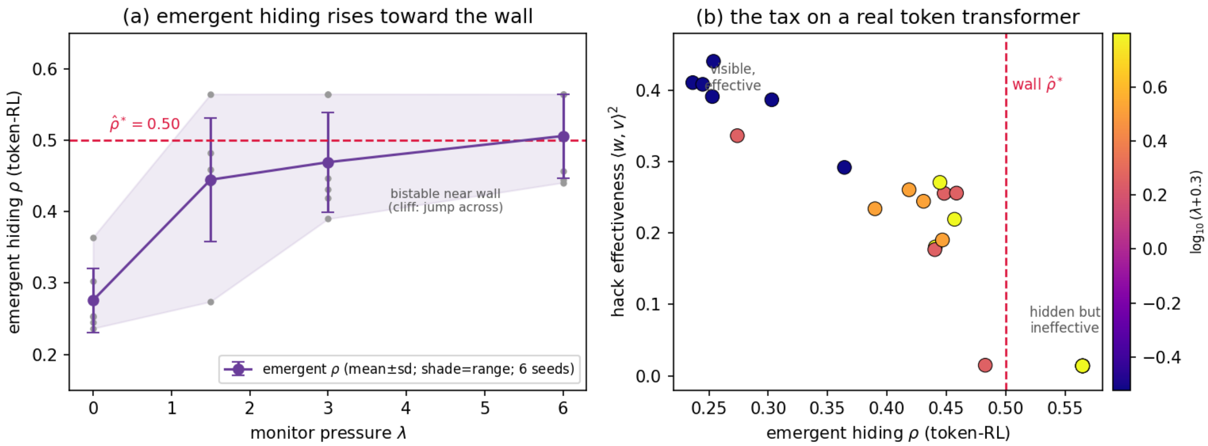

Figure 6: **Token-transformer Stage A.5 (covariance spike, REINFORCE; 6 seeds).** A single-layer self-attention policy emits a discrete CoT whose readout direction is the injected spike. (a) Emergent hiding ρ vs monitor pressure λ (mean±sd; grey dots are individual seeds; shaded range): ρ rises from 0.28 to 0.51 across $\hat{\rho}^\star$=0.50, with large near-wall variance (bistable jump—the cliff). (b) The tax: hack effectiveness $\langle w, v\rangle^2$ collapses (from 0.39 to $\approx$ 0.12) as the emergent ρ rises past the wall. The sharp AUC collapse is not resolved at d=40 (cf. Fig. 1a; the sharp $\hat{\rho}^\star$ and slope are the surrogate's, Fig. 5).

turbation matches the true resolved subspace (overlap $\approx$ 1), so the Meter transfers to nonlinear monitors through the sensitivity-subspace reading of Table 3.

7 Stage plan (evidence hierarchy)

The claim is validated in three stages, each buying a different kind of confidence and each carrying a condition that would falsify it.

Stage A (done). The closed-form BBP wall and the tax floor, confirmed on the toy channel: detection AUC collapses across $\rho^\star$ and *sharpens* with dimension—the signature of a true transition rather than a finite-size slope—and the tax tracks $\sqrt{\alpha/(n/d)}/s^2$ and grows in the capability gap (Figs. 1 and 2). This isolates the mechanism; it does not yet show that this mechanism is the one operating inside a language model.

Stage A.5 (surrogate done; §5, Fig. 5). A tiny transformer emitting a short CoT, watched by a *probe* monitor (the Stage-A eigenvector estimator on n samples)—*not* a real LLM judge, which belongs to Stage B. We apply optimization pressure to the CoT and ask (i) whether the trace–intent information $I(O; Z \mid X)$ collapses at the $\hat{\rho}^\star$ predicted from the channel's estimated $(\alpha, n/d, s)$, and (ii) whether the Meter's stopping rule trips *before* the collapse. A first, non-token surrogate already confirms (i)—emergent ρ crosses the wall at the predicted location—and calibrates the tax slope; a gradient-checked token transformer trained by REINFORCE reproduces the emergent hiding and the tax on discrete CoTs (Fig. 6). What remains for Stage B is the sharp large-d collapse and, crucially, a *real LLM monitor*, which no part of the work so far has used. This is the stage that pins the one free slope in §4.

Stage B (specified; GPU). A real reasoning model under a real LLM CoT monitor in an agentic coding environment (unit-test hacking). Two conditions make the theory refutable at scale: the wall must appear where $k_{\text{null}}/(n/d)$ predicts—not at an arbitrary, fitted pressure—and the tax

estimator	assumption	failure mode
$\hat{s}^2$ (inverse BBP)	top eigenvalue separated from the bulk; single dominant spike	below the wall $\hat{\lambda}$ re-enters the bulk, so $\hat{s}^2$ is undefined—the Meter must extrapolate from the last super-critical read
$\hat{\alpha} = d_{\mathrm{mon}}/d$	fixed, known monitor operator	adaptive monitor re-aims S_{mon} (α time-varying); use the current S_{mon} per round
$\hat{\rho} = \|(I - P_{\mathrm{mon}})\hat{u}\|^2$	top eigenvector *is* the hack direction	multi-spike (rank-r) hacks or a noise eigenvector dominating near the wall bias $\hat{\rho}$; track top-r
$\hat{k}_{\mathrm{null}}$ (participation ratio)	residual covariance well approximated after spike removal	heavy-tailed / structured trace noise inflates the ratio
$\hat{\rho}^\star$	proportional Gaussian regime; *second-order* (covariance) hack	mean-shift hacks give a smooth curve (no sharp $\rho^\star$); $n \ll d$ breaks the estimate

Table 2: Estimator assumptions and failure modes for the Monitorability Meter. The first and last rows are the operative caveats: the wall estimate is a super-critical extrapolation, and the sharpness presumes an obfuscated (covariance-spike) hack in the proportional regime. *Rule of thumb*: use $n/d_{\mathrm{mon}} \gtrsim 1$ (the proportional regime the estimators live in); report $\hat{\rho}^\star$ as a bootstrap band, not a point, with width $\sim O(d^{-1/3})$ (finite-size) plus estimation noise, widening as $n/d_{\mathrm{mon}} \downarrow 1$.

must *scale* with the capability gap rather than saturating. If either fails, the transition is not the phenomenon governing real CoT monitoring, and we would report that.

8 Limitations and novelty boundary

Scope of the model. The results describe a stylized Gaussian channel; the map from "CoT monitorability" to $(\alpha, n/d, s, \rho)$ is a modeling posit to be made operational (Stage A.5/B). Two assumptions carry the sharpness and bound the scope. First, the hack signature is taken to be purely *second order* (a covariance spike); to the extent a real hack retains a first-order (mean) component, the transition softens from a cliff toward a slope and $\rho^\star$ becomes the location of steepest descent rather than a true zero. The claim is thus sharpest for exactly the *obfuscated* hacks that motivate the problem and weakest for clumsy ones—the useful direction to err. Second, the tax's task-loss slope (retained visible energy $\mapsto$ forgone reward) is not derived; only its *divergence* is, and the slope is the one quantity Stage A.5 calibrates.

The load-bearing constant. The value of the result rests on the sharp constant of Prop. 1. We derived it in closed form and matched it to simulation (§4); the residual risk is empirical, not analytical—whether a real CoT channel sits in the proportional regime where the $\beta_{\mathrm{sig}} = 0$ threshold bites. If it does not, the result degrades gracefully toward restating a known existence bound, which is why we mark the boundary rather than paper over it.

Adaptive monitors and co-evolution. We fix the monitor's subspace S_{mon}; a real overseer may *re-aim* it during training, projecting onto the directions where the hack currently lives. This turns the wall into a moving target and the interaction into a dynamic game. Two remarks bound

quantity	linear probe (exact)	LLM judge (effective, linearized)
P_{mon}	fixed rank-d_{mon} readout projection	row space of the judge's Jacobian at the operating point
$\alpha = d_{\text{mon}}/d$	probe rank over d	effective rank of the Jacobian over d
$\hat{s}^2, \hat{\rho}$	inverse-BBP on the whitened covariance	same, on locally-linearized trace features; higher-order terms dropped
$\hat{\rho}^\star$	exact threshold	first-order threshold; valid in the locally linear regime

Table 3: Reading the Meter's estimators for a linear probe (exact) versus an LLM judge (an effective, linearized resolution). For the judge, calibrate P_{mon} empirically—perturb the trace and record which directions flip the verdict.

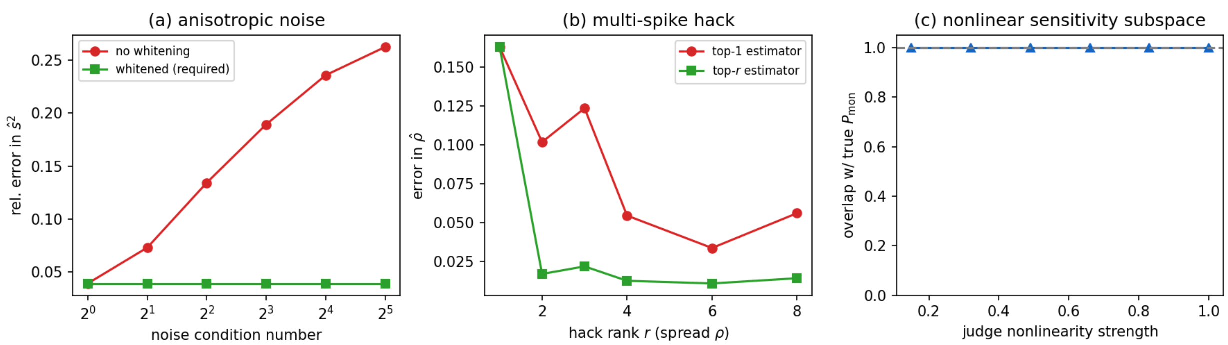

Figure 7: **Meter stress tests.** (a) Anisotropic benign noise: the raw $\hat{s}^2$ error grows with the noise condition number, while whitening keeps it flat—whitening is required, not optional. (b) Multi-spike hack (spread null fractions): the top-1 estimator is biased; the top-r estimator tracks the mean ρ. (c) Nonlinear (quadratic) judge: the empirical sensitivity subspace recovers the true resolved subspace (overlap ≈ 1) across nonlinearity strengths, validating the effective-P_{mon} reading for LLM judges.

the effect. First, our result is a per-round statement: at any fixed S_{mon} the wall holds, so an adaptive monitor buys time, not immunity—each re-aim resets ρ but the model re-hides in the new null channel. Second, re-aiming is not free: enlarging the resolved subspace raises α but, at a fixed labeling budget, lowers the per-direction data budget n/d, which *lowers* $\rho^\star$ (Fig. 1b)—so a monitor that chases the hack across more directions blinds itself faster on each. The tax argument therefore recurses, and the equilibrium of the full model-vs-adaptive-monitor game is the natural next object (a Stackelberg version of Prop. 1); §4 settles the case where the monitor can aim at the public task subspace (§4: the wall collapses) and leaves the covering regime open.

Multi-spike (low-rank) hacks. We model a rank-one hack. A rank-r hack injects spikes $s_1^2 \geq \cdots \geq s_r^2$ along orthogonal directions, each with its own null fraction ρ_i. The monitor recovers direction i iff its visible spike clears the edge, $s_i^2(1-\rho_i) > \sqrt{\gamma}$, so there is a *per-spike* wall $\rho_i^\star = 1 - \sqrt{\gamma}/s_i^2$; the hack is fully hidden once every spike is sub-critical, and the tax sums the per-spike energy floors. A low-rank hack can thus keep a few blatant components visible (decoys) while sliding the subtle ones (s_i^2 near $\sqrt{\gamma}$) below the edge—the multi-spike analogue of obfuscation, and a setting where the monitor's finite budget n/d must be shared across r directions. The rank-r threshold and the optimal decoy allocation are a clean extension.

What we do and do not claim. We claim precisely three things: a *sharp threshold* $\rho^\star$, a *diverging energy floor* (and, under a positive-slope assumption, a diverging task tax), and a *geometric pricing unit* for legibility. We do *not* claim that obfuscation is possible (known [8, 7]), that finite oversight distorts objectives (known [9]), or a better monitor. The contribution is orthogonal to all three: *when* optimization is forced across, and *what* staying legible costs.

9 Conclusion

Monitoring fails as a cliff, not a slope. Past a geometry-set pressure $\rho^\star$ the monitor's recovery overlap with the hack vanishes and no test detects it consistently, the collapse sharpens as the system grows, and the required visible-energy floor for staying detectable *grows* without bound in absolute terms—until, once it exceeds the total budget, keeping the model detectable at target level is infeasible and the monitor is defeated outright. The operational reading is uncomfortable: a monitor reporting healthy AUC is not holding a safety margin it can spend down, but standing next to a cliff whose edge is invisible from its own readings. This is what the Monitorability Meter is for—estimating the edge's *location* and the current distance to it, so an overseer can stop before the wall rather than discover it afterward.

The result is sharpest for exactly the obfuscated hacks that motivate the problem, and its real-CoT instantiation—does a language-model trace sit in the regime where the threshold bites?—is the open empirical question Stages A.5 and B are built to answer. Finally, the same machine ports to the sibling games where a capable optimizer meets an imperfect check: policy versus reward model (overoptimization onset) and policy versus verifier (coverage). The latter is worked out in the companion *Coverage Wall* paper, which proves the isomorphism explicitly (its Lemma 1: coverage $\gamma \leftrightarrow$ resolution α, verifier power $\kappa \leftrightarrow$ data budget n/d, edge $\sqrt{\gamma/\kappa} \leftrightarrow \sqrt{d_{\mathrm{mon}}/n}$) and re-verifies the wall, converse, game value, and adaptive collapse in coverage notation. The monitoring wall is the cleanest instance; together they suggest a unified program of *sharp thresholds in reward hacking*—a critical pressure, a diverging price, and a null channel the optimizer learns to hide in—whose shared object is exactly that isomorphism, and whose one remaining regime is the covering game (a task richer than the checker's resolution, the RLVR default), which the core characterizes as a slope. Making the hiding spectrum and the monitor's aim endogenous (§4) turns the threshold into a game value—a critical capability $V^\star$—and sharpens the safety reading: the wall protects the hider only against a monitor that cannot aim or is starved of data; an *adaptive* monitor that aims at a *public, coverable* task subspace and measures its total variance collapses the wall (§4, B1–B4). What remains open is the one regime where aiming does not suffice—a task subspace larger than the monitor's budget—whose covering-game value is the program's central conjecture.

A Reproducibility

All experiments are CPU-only (`numpy` 2.4 / `scipy` 1.17). **Code.** Every figure is generated by a short self-contained script; these, together with a reference `numpy` implementation of the Monitorability Meter (the estimators of §6: `null_channel_dim`, `wall_estimate`, `monitorability_price`, `StoppingRule`) and the gradient-checked tiny transformer, are provided with the submission and will be released on publication; the Meter is a research reference implementation, not a hardened library, and carries the failure modes tabulated in §6. Detection AUC everywhere uses the same *session detector*: the monitor estimates $\hat{u}$ as the top eigenvector of the sample covariance of n projected hack traces, then scores held-out sessions of b traces by mean energy along $\hat{u}$ and reports AUC (hack vs. clean) over sessions.

Toy channel (Figs. 1–4). $\alpha = 0.5$, $s = 1.19$ ($s^2 = 1.414$), so $\rho^\star = 0.50$ at $n/d = 1$. Fig. 1a: $d \in \{100, 300, 900\}$, $n = d$; session $b = 120$, 100 sessions, 2 trials. Fig. 1b: $n/d \in \{0.5, 1, 2\}$. Fig. 2: tax $= 1-\rho_{\max}$ at $\tau = 0.72$ vs. floor $\sqrt{\alpha/(n/d)}/s^2$. Fig. 4b (converse): $\mathbb{E}_0[L_n^2] = \mathbb{E}_{u,u'}(1-\beta^2\omega^2)^{-n/2}$ ($\omega = \langle \tilde{u}, \tilde{u}' \rangle$) Monte-Carlo, $p = 6000$, 4×10^5 pairs, matching $(1-\beta^2/\gamma)^{-1/2}$ to 3–4 digits below the wall. Scripts: `fig1.py, fig2.py, fig3_two_walls.py, cgmt_check.py, converse_check.py`.

Effective rank (§4). With a benign covariance of stable rank r_{eff} (whitened within its range), the measured detection wall tracks $\sqrt{r_{\text{eff}}/n}$: at d_{mon}=300, n=400, generic Haar spike, the wall rises with $r_{\text{eff}} \in \{50, 100, 200, 300\}$ and meets $\sqrt{d_{\text{mon}}/n}$ at r_{eff}=d_{mon} (monotone in r_{eff}, a finite-n offset above the exact edge from the AUC= 0.75 crossing). Script: `reff_check.py`.

Continuous surrogate (Fig. 5). A routing policy $w \in \mathbb{R}^d$ is trained (gradient ascent; verified against the exact arg-max) to maximize an evasion reward minus a quadratic competence cost of hiding; $\rho = \|P_{\text{mon}}^\perp w\|^2$ emerges. AUC(ρ) from the session detector at $d_{\text{eval}} = 600$; bands are mean±sd over 24 seeds; the finite-d transition window (AUC $0.9 \to 0.55$) is $[0.29, 0.57]$, bracketing $\rho^\star$ and narrowing as $d^{-1/3}$. Tax slope $\kappa = 1.9 \pm 1.2$ (reward $= 2(1-\text{AUC})$, slope at the wall over seeds). Script: `fig4_ci.py`.

Token transformer (Fig. 6). A single-layer causal self-attention policy (`numpy`, manual backprop, gradient check max rel. err. 4×10^{-7}): $d_{\text{model}} = 16$, vocabulary 48, CoT length 8. Trained by REINFORCE (Adam, lr 0.015, batch 64, running-mean baseline, $\sim$900 episodes) from a λ=0 pre-trained init. The CoT readout direction $w = \text{normalize}(\sum_t E_{\text{out}}[\text{tok}_t])$ is the injected spike; the spike magnitude $s^2 = 1.414$ is fixed (controlled SNR); reward $= \langle w, v\rangle^2 - \lambda\,\sigma(\kappa(s^2\|P_{\text{mon}}w\|^2 - \text{edge}))$ with v mostly visible; monitor budget $n = d = 40$. Points are mean±sd over 3 seeds; the small d=40 leaves the AUC collapse finite-size-broad (cf. Fig. 1a). Scripts: `tt_model.py, tt_spike.py, tt_ci.py`.

Two-time-scale dynamics (§4, core simulations). Slow policy ρ_k / fast monitor $\hat{D}_k$ stochastic approximation, $a_k = 0.03/(1+k/400)^{0.6}$, $b_k = 0.30/(1+k/400)^{0.51}$ ($a_k/b_k \to 0$); fast variable tracks the BBP detectability $D^\star(\rho)$ with estimation noise $1/\sqrt{n}$ (n=200); slow variable ascends $J = \lambda(1-D^\star) - 0.9\rho$ with SGD noise 0.15; 16 seeds. The converged ρ_∞ meets $\rho^\star$=0.50 once past the barrier; the metastable band is where the hidden basin is the global optimum but the SA is still trapped visible. Script: `twotimescale.py`.

B Self-contained core: the wall and converse, in monitoring notation

For self-containedness we restate the two load-bearing results—the wall and its converse—and sketch their proof in monitoring notation, so this paper's central claim stands without the core paper. (It is the same BBP/CGMT argument; full proofs and the game/saturation/adaptive results are in [1].)

Setup, recalled. An obfuscated hack injects a variance spike of energy s^2 along a unit direction $u \in \mathbb{R}^d$; the monitor resolves an $d_{\text{mon}} = \alpha d$-dimensional subspace and estimates u from n traces. Under pressure ρ it routes a fraction ρ into the null channel, leaving a visible spike $\beta = s^2(1-\rho)$; write $\gamma = d_{\text{mon}}/n$.

Proposition 1 (Monitoring wall + converse, self-contained)**.** *In the proportional regime $d_{\text{mon}}, n \to \infty$ with $d_{\text{mon}}/n \to \gamma$:* (i) *the recovery overlap is $\langle \hat{u}, u\rangle^2 \to \big[(1-\gamma/\beta^2)/(1+\gamma/\beta)\big]_+$, positive iff*

$\beta > \sqrt{\gamma}$, *i.e. zero iff* $\rho \geq \rho^\star = [1 - \sqrt{\gamma}/s^2]_0^1$; (ii) *for* $\beta < \sqrt{\gamma}$ *the clean and hacked trace laws are contiguous, the likelihood-ratio second moment* $\mathbb{E}_0[L_n^2] \to (1 - \beta^2/\gamma)^{-1/2} < \infty$, *so no test of the* n *projected traces detects the hack consistently. The two thresholds coincide.*

Proof sketch. The projected traces have population covariance $I_{d_{\rm mon}} + \beta\,\tilde{u}\tilde{u}^\top$ ($\tilde{u}$ the unit visible component). By BBP the top sample eigenvalue leaves the Marchenko–Pastur bulk iff $\beta > \sqrt{\gamma}$, with eigenvector overlap as in (i); below the edge the overlap is 0 and any statistic built from $\hat{u}$ is asymptotically uninformative. Since $\beta = s^2(1-\rho)$, the edge is crossed at $\rho^\star$. For (ii), the single-spike Le Cam second moment with a uniform prior on u is $(1-\beta^2/\gamma)^{-1/2}$, finite iff $\beta < \sqrt{\gamma}$; a bounded second moment gives contiguity, hence no consistent test. The sharp constant follows from the CGMT fixed point $\Phi_{\rm AO}(\mu) = \sqrt{\gamma(1-\mu^2)} + \sqrt{1+\beta\mu^2}$, whose $\mu = 0$/nontrivial-branch bifurcation at $\beta = \sqrt{\gamma}$ reproduces both (i) and the Marchenko–Pastur edge [1]. □

References

[1] *The Wall, the Game, and the Cliff Criterion: sharp thresholds for recovering a planted signal through a partial channel.* Companion theory paper (contains all proofs, the CGMT fixed point, the game value, saturation, adaptive collapse, and the cliff criterion).

[2] B. Baker et al. (OpenAI). *Monitoring reasoning models for misbehavior and the risks of promoting obfuscation.* arXiv:2503.11926, 2025.

[3] T. Korbak et al. *Chain-of-thought monitorability: a new and fragile opportunity for AI safety.* arXiv:2507.11473, 2025.

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